
Fusioning Image and Language Understanding

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Submitted in partial fulfillment of the requirements
of the degree of Bachelor of Science in Computer Science and Engineering

November 4, 2024



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Abstract

In academic institutions, the manual handling of various documents, including exam scripts, administrative records, and evaluations, is often labor-intensive and prone to human error. Traditional digitization systems frequently struggle with the complexities presented by mixed handwritten text, layouts, and variations in document structures. Furthermore, annotated datasets may not be available due to privacy concerns, particularly in university evaluations involving sensitive information making it difficult for the model to learn effectively. To address these challenges, we propose uFOIL, an ensemble-based unsupervised framework that utilizes the fusion of image processing and language processing to automate the extraction and validation of key information. The proposed framework utilizes a majority voting mechanism that integrates four state-of-the-art optical character recognition systems. Additionally, a transformer architecture is introduced to further enhance the contextual understanding and structured formatting of the extracted text. The model achieves outstanding performance, recording 95.77%, and 96.48% accuracy for student names, and IDs, respectively based on a dataset of 412 exam script samples. Additionally, it demonstrates an accuracy of 95.07% in total mark validation.

Acknowledgements

First and foremost, we express our heartfelt gratitude, and praise to the Almighty, and Graceful Allah for granting us the strength, wisdom, and perseverance to complete this research.

This work would not have been possible without the input and support of many individuals over the last two trimesters. We extend our sincere appreciation to our thesis supervisor, Prof. Dr. Md. Motaharul Islam, and our course teacher, Prof. Dr. Hasan Sarwar, for their invaluable guidance and support throughout this project. Their expertise and insights have been instrumental in shaping our ideas and methodologies.

Lastly, we owe a deep debt of gratitude to our families, including our parents, for their unconditional love, courage, and immense emotional support.

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Chapter 1

Introduction

This chapter provides the necessary background analysis, motivation, and objectives for our proposed framework. It also includes major contributions and the structure of the report.

1.1 Background

The automation of document processing has revolutionized various sectors, including education. Recently, educational institutions have increasingly relied on technology to simplify administrative tasks, enhance efficiency, and improve decision-making processes [1, 2, 3, 4].

Universities and other academic institutions are responsible for managing vast amounts of data throughout the academic cycle, from student enrollment to examination processing. One of the most resource-intensive tasks in this process is handling physical exam scripts, which are central to most academic evaluations. As institutions seek to improve operational efficiency and reduce human errors, the demand for automation in document processing has grown significantly [5]. Historically, processing exam scripts has involved substantial manual effort in most academic institutions [5]. Administrative staff must extract information, verify it against academic records, and ensure accurate processing for each student. This labor-intensive and time-consuming process is susceptible to human error, particularly in large institutions handling thousands of scripts each semester [6].

Recent advances in image processing, particularly in Optical Character Recognition (OCR) and document analysis techniques, have opened new doors for automating these processes [7, 8, 9]. OCR technologies have evolved from recognizing printed text to more sophisticated methods that can interpret handwritten content and deal with complex document layouts [10, 11]. Despite these developments, challenges remain, particularly with varied handwriting styles, mixed document layouts, and non-standardized formatting, which traditional OCR systems still struggle to address effectively.

In addition to OCR, Natural Language Processing (NLP) has emerged as an essential component in the analysis and interpretation of textual data across various fields. NLP technologies facilitate tasks such as information extraction [12, 13], content classification

[14, 15], and sentiment analysis [16] which enables machines to process and understand human language [17].

Moreover, advancements in Machine Learning (ML) and Artificial Intelligence (AI) have begun to transform the landscape of document processing. Techniques such as Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) have shown promise in improving image analysis and augmenting training datasets [18, 19, 20]. Additionally, integrating these AI technologies with data mining and knowledge management systems can facilitate intelligent decision-making processes. As educational institutions increasingly adopt these technologies, the interplay between OCR, NLP, and machine learning presents a unique opportunity to automate the handling of exam scripts.

1.2 Motivation

Exam control departments and instructors have traditionally invested significant effort into analyzing and evaluating handwritten exam scripts. This manual grading process is time-consuming and adds considerable burdens to educational institutions. Beyond the time it requires, hand-grading also introduces the risk of human error, especially when it comes to scoring, totaling, and organizing data for documentation. Small errors can accumulate and compromise the accuracy of final assessments. Simplifying this grading process has become essential, especially as educational institutions handle larger numbers of exams and strive for efficiency. Automating the entire grading pipeline could save time, reduce human error, and increase the reliability of exam results. Handwritten text, however, presents unique challenges for automated systems. Traditional OCR technology struggles with varied handwriting styles, inconsistent formatting, and different layouts. This makes it difficult to accurately capture and process handwritten content in exam scripts. Additionally, exam scripts often include both handwritten and printed text, as well as tables for organizing scores, which adds further complexity.

Our motivation is to address these challenges by developing a system that can process handwritten and printed text reliably, handle different document layouts, and ensure accurate data extraction. This would give educators a dependable way to process exams quickly and with fewer errors freeing up time for teaching and student engagement.

1.3 Objectives

The main objectives of this project are:

1. To develop a system that can digitize and process handwritten exam scripts.
2. To accurately extract text from scripts, handling different handwriting styles and layouts.
3. To extract key information, such as student name, ID, and scores, in a structured format.

4. To automate the calculation of final grades from extracted marks.
5. To improve the speed and accuracy of grading and reduce time and effort for exam evaluation.
6. To create a scalable solution that can process large numbers of exam scripts efficiently.

1.4 Project Overview and Major Contributions

Despite advancements in automation, current systems still struggle to handle complex documents characterized by mixed content types, including structured fields, handwritten data, and irregular layouts. These challenges necessitate a more effective approach to ensure reliable information extraction. Thus, in this study, we propose a comprehensive framework that incorporates image preprocessing, segmentation, and text extraction through an ensemble of advanced OCR techniques, alongside Natural Language Processing methods and GAN-based data augmentation. Our framework operates in an unsupervised manner which allows the adaptation of varying document structures without the need for annotated data, thus enhancing its applicability in real-world scenarios where such datasets may be confidential. The major contributions of our study are as follows:

- A comprehensive preprocessing pipeline is developed, including dynamic contrast adjustment, text-specific rotation correction, and denoising techniques, to produce high-quality images for optimal text extraction.
- A novel unsupervised segmentation framework is implemented, supplemented by a GAN-based data augmentation approach that generates realistic variations of exam scripts addressing the limitations of conventional cropping and rotation techniques.
- An ensemble of state-of-the-art OCR models is developed for text extraction to effectively handle both printed and handwritten content.
- A post-processing strategy is introduced which features a confidence scoring mechanism to enhance the accuracy of extracted characters and format-based validation techniques to correct recognition errors.
- A Transformer-based architecture is integrated into the output text formation to enable dynamic contextual understanding and structured formatting of the extracted text.

1.5 Methodology

In this study, we propose an ensemble-based unsupervised learning framework, uFOIL. Our framework employs a systematic approach to enhance information extraction from

exam scripts. We start by preprocessing the input images to improve their quality through contrast adjustment, noise reduction, and correction of any misalignments or rotations. This preprocessing is crucial for ensuring that the subsequent extraction processes work with clear and consistent data. Next, we split the images into two sections: one for student information and another for marks tables, allowing us to focus our extraction efforts more accurately. We utilize a combination of Optical Character Recognition (OCR) techniques that can handle both handwritten and typed text, accommodating the various writing styles found in exam scripts. After extracting the text, we standardize the data for easier analysis and validate the extracted marks against the total marks indicated in the image filenames. This validation is essential for ensuring the reliability of our system in real educational contexts. Finally, we evaluate the framework's performance by measuring its effectiveness in accurately extracting names, student IDs, and marks, ensuring that our solution meets the needs of educational institutions.

1.6 Preliminaries

In order to understand the concepts and techniques discussed in this report, readers should have foundational knowledge in several key areas. This section briefly introduces the key areas essential to understand this report.

- **Image Processing:** Image processing techniques are fundamental for enhancing and preparing images for analysis. Core techniques such as contrast adjustment, noise reduction, and rotation correction are often applied to improve the quality and readability of scanned or photographed documents. These preprocessing steps play a vital role in ensuring that subsequent stages of analysis can work with clear and standardized input images.
- **Optical Character Recognition (OCR):** OCR technology is essential for extracting text from images, enabling the transformation of printed or handwritten text into digital formats. OCR models analyze images to detect characters and words, often requiring sophisticated algorithms to handle varying fonts, handwriting styles, and text orientations.
- **Natural Language Processing (NLP):** NLP encompasses a variety of techniques used to process, interpret, and manipulate text data. Common tasks in NLP include tokenization, language modeling, and syntactic analysis. By allowing machines to understand and structure human language, NLP techniques are particularly useful in fields where extracting specific information from unstructured text is required.
- **Transformer Models:** Transformers are a type of neural network architecture widely used in NLP due to their ability to model long-range dependencies in text data. Through an attention mechanism, Transformers can focus on different parts of a text sequence, capturing contextual relationships between words.

- **Generative Adversarial Networks (GANs):** GANs are a class of machine learning models used to generate realistic synthetic data. Comprised of a generator and a discriminator, GANs learn to produce data that closely resembles the training set by competing in an adversarial process. GAN-based augmentation can be especially useful when working with limited datasets, as it allows for the creation of diverse, realistic samples.

This foundational knowledge will help readers better understand the methods and techniques discussed in the subsequent sections of this report.

1.7 Organization of the Report

The structure of the report is as follows:

- **Introduction:** Provides an overview of the project, including objectives, motivation, project overview, and major contributions.
- **Related Works:** Reviews relevant research, and identifying gaps that our project seeks to address.
- **Project Design:** Details the system design and implementation including both functional and non-functional requirements.
- **Result Analysis:** Details the experimental setup, results, and outcome of our proposed system.
- **Standards and Constraints:** Discusses standards followed (software, hardware, communication) and considers design constraints, including economic, and ethical factors.
- **Conclusion:** Summarizes the project highlighting key contributions, limitations, and areas for future improvement.

Chapter 2

Related Works

In this section, we have presented the essential background information, reviewed existing applications and relevant research, and identified gaps in the current literature. Figure 2.1 provides a summary of the explored works.

2.1 Literature Review

Recent advancements in table detection and structure recognition have significantly improved the ability to extract tabular data from a range of document types, from structured reports to unstructured images. Various models [21, 22, 23, 24, 25, 26] have leveraged deep learning techniques to locate tables and identify cell structures within complex layouts accurately. For instance, Agarwal et al. [21] developed CDeC-Net, an end-to-end architecture that employs a multi-stage Cascade Mask R-CNN framework. This model progressively refines table proposals by dynamically adjusting intersection over union (IoU) thresholds, enabling high precision in distinguishing table boundaries. Similarly, Paliwal et al. [22] proposed TableNet, a model using the VGG19 backbone for multi-task learning, which jointly detects table regions and cell boundaries to capture the complex layout within tabular structures better. Another notable model, CascadeTabNet by Prasad et al. [23], integrates table detection and cell segmentation into a single model. By utilizing the High-Resolution Network (HRNet), CascadeTabNet ensures high accuracy in table structure delineation, particularly in densely populated tables. Qiao et al. [24] presented Local and Global Pyramid Mask Alignment (LGPM), which tackles complex table structures through a dual-branch approach, achieving both local binary segmentation and global cell prediction to detect empty cells. Riba et al. [25] introduced a graph-based neural network framework, effectively mapping table structures in documents with intricate layouts. Finally, Zacharias et al. [26] used traditional image processing combined with deep learning to detect tables from images, offering an alternative to purely deep learning-based approaches for table extraction in low-resource settings.

Recent years have also seen the rise of transformer-based architectures [27, 28, 29] in document understanding, enabling seamless integration of multi-modal data, such as text,

visual, and spatial features. Appalaraju et al. [27] introduced DocFormer, a transformer model designed for Visual Document Understanding (VDU), which leverages self-attention mechanisms to process complex document layouts by combining various feature types. This model has shown efficacy in handling a wide range of document formats by integrating spatial relationships directly into its processing pipeline. Similarly, Li et al. [28] developed an unsupervised image-sentence matching model using transformers to create graph representations. This approach enhances cross-modal feature learning, allowing for more robust document understanding, even in complex layouts. Li et al. [29] further expanded transformer use in document processing with DiT (Document Image Transformer), a self-supervised model trained on various document tasks. DiT’s pre-training enables effective document layout analysis and even table detection, showcasing the versatility of transformers in multi-task document understanding. Tewel et al. [30] extended this work by developing ZeroCap, a zero-shot visual-semantic matching model that leverages large language models to generate descriptive captions from images, further demonstrating the transformer’s potential in complex document interpretation.

For tasks beyond table detection, several studies [31, 32, 33, 34, 35] have developed models focused on specific object and element detection, which is crucial for processing diverse document types. Jiang et al. [31] created TabCellNet, which incorporates a Hybrid Task Cascade (HTC) architecture, combining object detection and instance segmentation for accurate cell recognition. TabCellNet refines its detection outputs using Soft-Non-Maximum-Suppression (Soft-NMS), which improves the accuracy of tabular element localization. Ma et al. [32] proposed a model for chart data extraction that uses a Cascade R-CNN in combination with a deep learning-based element detection module. This model is particularly useful in applications requiring high-precision charts and graphical element detection. Lertsawatwicha et al. [33] applied deep learning along with Tesseract OCR to automate lot number detection in medical images, demonstrating the use of OCR in highly specific document applications. Lu et al. [34] introduced a multi-scale feature extraction approach within Visual Question Answering (VQA) systems, combining various levels of text representation for more nuanced text understanding. Zhu et al. [35] presented DocBed, which employs Fully Convolutional Networks (FCNs) with Mask R-CNN to segment complex layouts, making it especially useful for intricate documents like newspaper layouts.

In addition to deep learning approaches, heuristic and rule-based techniques remain integral to information extraction and document analysis, particularly in resource-limited or high-specificity contexts. Milosevic et al. [36] employed a hybrid method, combining heuristic rules with machine learning to extract biomedical information from tables by applying predefined rules for cell classification and tagging. Grisci et al. [37] developed a relevance aggregation approach to enhance feature selection for neural networks, improving table data relevance estimation. Holecek et al. [38] employed a graph-based approach for invoice table extraction, using graph convolutions to assign labels to text boxes in structured table layouts. Retsinas et al. [39] proposed a keyword-spotting technique for

handwritten documents that avoids supervised training by employing contrast normalization and spatial descriptors for feature extraction. In historical document processing, Zagoris et al. [40] focused on word spotting in handwritten texts, applying local features to improve word recognition without training data. Lastly, Zhang et al. [41] proposed MDERank, an unsupervised keyphrase extraction model that uses a masking strategy to identify keyphrases across long-form documents effectively.

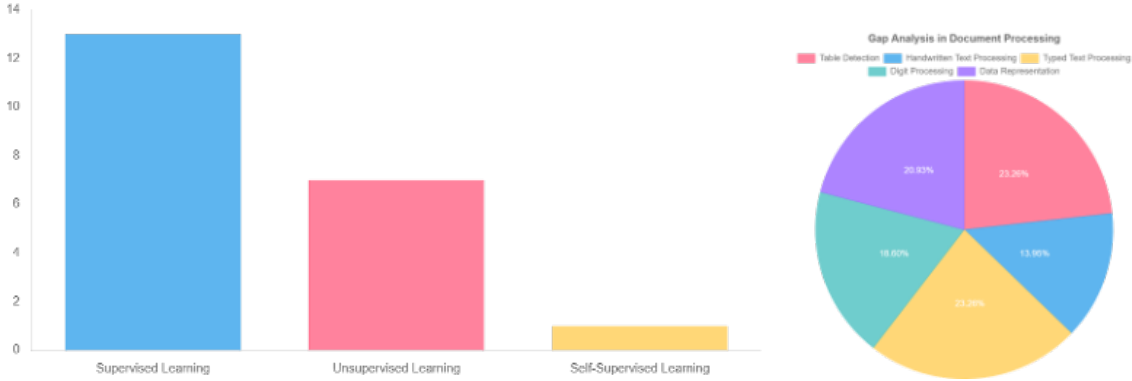


Figure 2.1: Overall summary of the related works

2.2 Gap Analysis

This section identifies key gaps in the literature on table detection, document understanding, and text processing techniques, as outlined in Table 2.1.

Despite significant advances in table detection and structure recognition, existing models often focus on individual subtasks rather than providing a comprehensive solution for end-to-end document processing. Particularly, there remains a lack of robust, unsupervised models capable of processing both handwritten and typed text, as well as digit recognition within complex tabular data.

Many existing models, such as CDeC-Net by Agarwal et al. [21] and CascadeTabNet by Prasad et al. [23], successfully address table detection but primarily target typed text and rely on supervised learning, necessitating large annotated datasets. Although effective in structured environments, these models struggle with the variability and noise present in real-world applications where handwritten elements are common, such as exam scripts, invoices, or historical documents. Moreover, while models like DocFormer by Appalaraju et al. [27] and DiT by Li et al. [29] utilize transformer-based architectures for multi-modal document understanding, their scope is often limited to general document layout analysis rather than focused table detection. They also lack specific mechanisms to seamlessly integrate multi-type text processing and unified data representation, which are crucial for consistency across various document formats.

Efforts like TabCellNet [31] and LGPMA [24] have addressed digit recognition alongside typed text processing, but these solutions often omit the handwritten text and rely on

a supervised approach, limiting their adaptability to new data. While some unsupervised methods, such as those by Zagoris et al. [40] and Holecek et al. [38], have focused on detecting text without labeled data, these techniques are often tailored to specific document types or employ traditional OCR systems like Tesseract, which struggle with low-quality inputs, particularly handwritten ones. Additionally, existing unsupervised methods generally lack a unified data representation mechanism that can consolidate various formats, including text and numerical values, into a cohesive data structure.

Our proposed framework, uFOIL, directly addresses these limitations by integrating multiple capabilities into a single, unsupervised solution. Specifically, uFOIL combines robust table detection, handwritten and typed text processing, digit recognition, and unified data representation. By eliminating the reliance on annotated datasets, uFOIL offers a scalable solution adaptable to diverse document types and real-world conditions.

Table 2.1: Gap Analysis

Ref	Approach	Table Detection	De- Text Processing	Handwritten Text Processing	Typed Text Process- ing	Digit Process- ing	Data Rep- resentation
[21]	Supervised	Yes		No	No	No	No
[27]	Unsupervised	No		Yes	Yes	Yes	Yes
[31]	Supervised	Yes		No	Yes	Yes	No
[32]	Supervised	No		No	No	No	No
[22]	Supervised	Yes		No	Yes	No	No
[23]	Supervised	Yes		No	No	No	No
[24]	Supervised	Yes		No	Yes	No	No
[37]	Supervised	Yes		No	Yes	No	No
[25]	Supervised	Yes		No	Yes	No	No
[33]	Unsupervised	No		No	Yes	Yes	Yes
[28]	Unsupervised	No		No	Yes	Yes	Yes
[34]	Supervised	No		No	Yes	Yes	Yes
[39]	Supervised	No		Yes	No	No	Yes
[30]	Supervised	No		No	Yes	Yes	Yes
[26]	Unsupervised	No		No	Yes	Yes	No
[40]	Unsupervised	No		Yes	No	No	No
[41]	Unsupervised	No		No	Yes	Yes	Yes
[35]	Supervised	Yes		No	Yes	No	Yes
[29]	Self-supervised	Yes		No	Yes	No	No
[38]	Unsupervised	Yes		No	Yes	No	No
[36]	Supervised	Yes		No	Yes	Yes	Yes
Ours	Unsupervised	Yes		Yes	Yes	Yes	Yes

Chapter 3

Project Design

This chapter outlines the project design, detailing the requirements analysis, including functional and non-functional requirements, project planning, task allocation, and a comprehensive overview of the implementation process.

3.1 Requirement Analysis

3.1.1 Functional Requirements

The following functional requirements outline the core capabilities and objectives of our research:

1. **Image Processing:** Design algorithms to accurately extract the handwritten text from exam scripts in various image formats, accounting for factors like resolution, lighting conditions, and handwriting variations.
2. **Preprocessing Pipeline:** Implement preprocessing techniques like contrast adjustment, rotation correction, noise reduction, and picture enhancement to improve the quality of the retrieved text data.
3. **Document Segmentation:** Design algorithms that separate exam scripts from any background elements and detect structural components such as headers, footers, paragraphs, and tables accurately.
4. **Table Parsing:** Develop methods for reliably extracting table structures from test documents, including table boundaries, column headers, and cell contents.
5. **Text-to-Structured Representation:** Create methods to convert extracted text into a structured format that resembles an exam script, containing metadata such as question numbers, marks, student IDs, and time stamps.
6. **Model Integration:** Ensure that OCR and NLP models are seamlessly integrated to process extracted text and generate smooth representations that allow components to interact.

7. **Ensemble Learning:** Explore ensemble learning techniques to aggregate outputs from many OCR and NLP models, thereby improving accuracy and dependability.
8. **Transformer Model Implementation:** Develop custom transformer-based NLP models tailored to the specific requirements of exam script analysis, capturing contextual relationships effectively.
9. **Document Layout Analysis:** Design algorithms for layout analysis to identify and classify different regions within exam scripts, facilitating segmentation and component extraction.
10. **Text Representation and Encoding:** Design encoding schemes to represent textual information in a structured format, making it suitable for downstream tasks and including metadata for comprehensive analysis.

3.1.2 Nonfunctional Requirements

The following are the nonfunctional requirements of this research:

1. **Accuracy:** To enable dependable grade computation and data extraction, the system's text extraction and recognition must be highly accurate, particularly for handwritten writing.
2. **Efficiency:** The system must process exam scripts rapidly, reducing the time necessary for grading and calculating results, hence increasing educational institutions' production.
3. **Reliability:** The system should function consistently under a variety of settings, including differences in image quality, handwriting styles, and document layouts while maintaining a low error rate in extracted data.
4. **Robustness:** The system should be able to manage image distortions, noise, and rotation with minimum influence on data extraction accuracy, regardless of image quality variations.

3.2 Project Task Allocation

For our project task allocation, we have identified major tasks which are outlined below:

1. Define Research Problem:
 - 1.1 Refine research objectives and scope based on identified challenges
 - 1.2 Identify key challenges in text extraction from exam scripts
 - 1.3 Finalize research problem statement and objectives

2. Explore Existing Solutions:
 - 2.1 Review existing techniques for text extraction
 - 2.2 Evaluate the performance and limitations of current methods
 - 2.3 Identify potential gaps and opportunities for improvement
3. Dataset Acquisition:
 - 3.1 Collect exam script datasets, considering variations in handwriting and document formats.
 - 3.2 Preprocess exam script data for training and evaluation.
 - 3.3 Ensure data quality and integrity through proper validation and cleaning processes.
4. Development of Preprocessing Pipeline:
 - 4.1 Implement algorithms for contrast adjustment, rotation normalization, and noise reduction
 - 4.2 Design techniques for document segmentation and table parsing
 - 4.3 Optimize the preprocessing pipeline for efficiency and accuracy in text extraction
5. Integration of OCR and NLP Models:
 - 5.1 Select and configure OCR and NLP models suitable for text extraction from exam scripts
 - 5.2 Develop mechanisms for model integration and information fusion
 - 5.3 Explore ensemble learning approaches to leverage the strengths of multiple models
6. Transformer Model Implementation:
 - 6.1 Research and understand the architecture and functioning of transformer-based models
 - 6.2 Adapt the model for the specific requirements script analysis
 - 6.3 Train and fine-tune the model on exam script datasets for text representation and encoding
7. System Testing and Evaluation:
 - 7.1 Design experiments to evaluate the accuracy, efficiency, and robustness of the developed system.
 - 7.2 Conduct comprehensive testing across different scenarios.
 - 7.3 Analyze and interpret experimental results to assess the performance of the system.
8. Documentation and Reporting:
 - 8.1 Prepare detailed documentation outlining the research methodology, findings, and conclusions
 - 8.2 Write research papers and reports for publication

This task assignment section outlines the specific tasks and activities to be undertaken in each phase of the research project. Table 3.1 highlights tasks allocation of our project:

3.3 Project Workflow

Our uFOIL framework employs a systematic approach to enhance information extraction from exam scripts. We start by preprocessing the input images to improve their quality through contrast adjustment, noise reduction, and correction of any misalignments or rotations. This preprocessing is crucial for ensuring that the subsequent extraction processes work with clear and consistent data. Next, we split the images into two sections: one for student information and another for marks tables, allowing us to focus our extraction efforts more accurately. We utilize a combination of Optical Character Recognition (OCR) techniques that can handle both handwritten and typed text, accommodating the various writing styles found in exam scripts. After extracting the text, we standardize the data for easier analysis and validate the extracted marks against the total marks indicated in the image filenames. This validation is essential for ensuring the reliability of our system in real educational contexts. Finally, we evaluate the framework’s performance by measuring its effectiveness in accurately extracting names, student IDs, and marks, ensuring that our solution meets the needs of educational institutions. Fig. 3.1 illustrates the overall pipeline of our methodology.

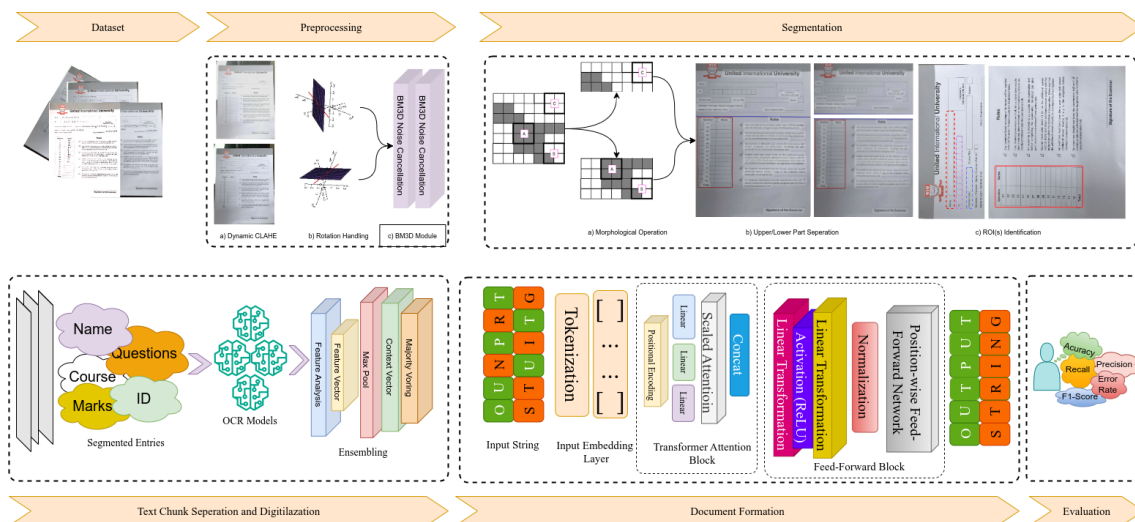


Figure 3.1: Overall pipeline of the proposed methodology

3.4 Dataset

The dataset used in this study consists of 412 custom-created exam script samples based on a standardized exam script template from United International University, Bangladesh. The template was obtained with the necessary permission, adhering to the institution's

data privacy and ethics guidelines. We populated this template with diverse entries to create a realistic dataset that simulates actual examination scripts. The images are taken using smartphone cameras in variable format (i.e., JPG, PNG, etc.) under varied conditions, which resulted in differences in rotation and image quality. As visualized in Fig. 3.2, each exam script contains two key sections. The upper part includes fields such as the student's name, ID, course name, course code, respective trimester, section name, and exam invigilator's signature. For each label, there is a designated box where the corresponding information is provided. The lower part, on the other hand, contains the examiner's signature and marks entries for individual questions recorded in a table format, listing the marks obtained for each question. The summation of total marks is not explicitly provided within the table; instead, they are derived manually for validation. For this study, we have focused on the following fields: students' names, IDs, course codes, and question-mark pairs evaluations. The dataset distribution is shown in Table 3.2. To further enhance the dataset size and improve model performance, we have augmented the dataset to a total of 712 samples using GAN-based augmentation techniques, which generated realistic variations while preserving the structural integrity of the exam scripts. More details about this augmentation process are discussed in Section 3.5.4.

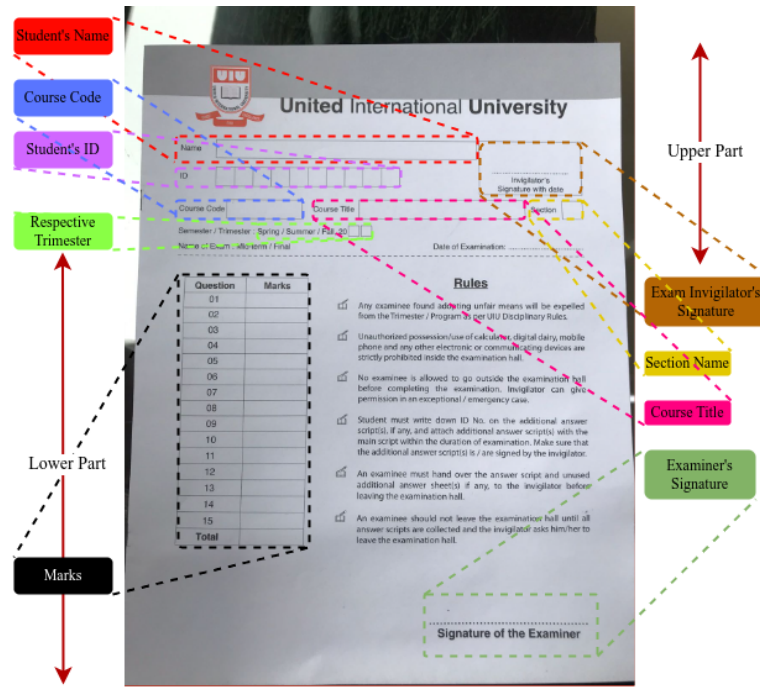


Figure 3.2: Key components of dataset sample

3.5 Preprocessing

To achieve optimal accuracy in extracting textual information from images of academic answer scripts, we have implemented a comprehensive preprocessing pipeline which con-

sists of four steps such as dynamic contrast adjustment, precise rotation correction, advanced image enhancement, and GAN-based augmentation techniques. The preprocessing pipeline for exam script images is outlined in Algorithm 1. The steps are elaborated in the subsequent sections.

Algorithm 1 Preprocessing pipeline

```

1: Input: Exam Script Images  $I = \{i_1, \dots, i_n\}$ 
2: Key Parameters: CLAHE,  $\theta$ , OCR, BM3D, GAN,  $I'$ 
3:
4: for all images  $i$  in  $I$  do
5:    $I_{\text{CLAHE}} \leftarrow \text{apply\_CLAHE}(i)$ 
6:    $I_{\text{edges}} \leftarrow \text{edge\_preserving\_filter}(I_{\text{CLAHE}})$ 
7:    $\theta \leftarrow \text{detect\_skew}(I_{\text{edges}})$ 
8:    $I_{\text{rotated}} \leftarrow \text{rotate\_image}(I_{\text{edges}}, \theta)$ 
9:    $I_{\text{denoised}} \leftarrow \text{apply\_BM3D}(I_{\text{rotated}})$ 
10:   $I' \leftarrow \{I' \cup I_{\text{denoised}}\}$ 
11: end for
12:
13: Step 1: Contrast Adjustment
14: for all images  $i'$  in  $I'$  do
15:    $I_{\text{CLAHE}} \leftarrow \text{apply\_dynamic\_CLAHE}(i')$ 
16: end for
17:
18: Step 2: Text Region-Based Rotation
19: for all images  $i'$  in  $I'$  do
20:    $\theta_{\text{avg}} \leftarrow \text{compute\_average\_text\_region\_angle}(i')$ 
21:    $I_{\text{aligned}} \leftarrow \text{refine\_rotation}(i', \theta_{\text{avg}})$ 
22: end for
23:
24: Step 3: Image Denoising
25: for all images  $i'$  in  $I'$  do
26:    $I_{\text{final}} \leftarrow \text{apply\_BM3D}(i')$ 
27: end for
28:
29: Step 4: GAN-Based Augmentation
30: Train GAN with generator  $G$  and discriminator  $D$ 
31: for all images  $i'$  in  $I_{\text{final}}$  do
32:    $I_{\text{synthetic}} \leftarrow \text{GAN\_generate}(G, D, i')$ 
33:    $I_{\text{augmented}} \leftarrow I_{\text{augmented}} \cup I_{\text{synthetic}}$ 
34: end for
35:
36: Output: Preprocessed and Augmented Images
  
```

3.5.1 Contrast Adjustment

Poor contrast and uneven illumination of samples affect the readability of important details. Our dataset contains samples with varying text densities, inconsistent lighting,

and handwritten markings, which pose additional challenges in accurately distinguishing characters and numbers.

To address these issues, we have employed a modified approach of the Contrast Limited Adaptive Histogram Equalization (CLAHE) method [42], which enhances contrast by operating on small regions (tiles) of the image. CLAHE limits contrast enhancement to prevent noise amplification, using a clipping threshold to control histogram redistribution in each local region. The standard CLAHE formula for adjusting pixel intensity $I(x, y)$ is given by:

$$I_{CLAHE}(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \times (L - 1), \quad (3.1)$$

where I_{min} and I_{max} are the minimum and maximum intensities within the local region, and L is the number of intensity levels.

However, traditional CLAHE can be suboptimal for our application due to the diverse nature of the samples. To enhance performance, we have modified the existing CLAHE approach by introducing a dynamic tile size adjustment based on local text density. We have analyzed image regions for text density and adapted the tile size accordingly, using smaller tiles in denser text areas and larger tiles in sparser regions. Additionally, the clipping limit is dynamically adjusted based on the overall brightness and variance of the image. In bright regions, the clipping limit is reduced to prevent over-enhancement, while in darker regions, a higher limit is applied to increase visibility. The modified CLAHE intensity adjustment is expressed as:

$$I_{modified}(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \times (L - 1) \times \alpha(x, y), \quad (3.2)$$

where $\alpha(x, y)$ is a scaling factor derived from local brightness and density metrics, allowing for adaptive contrast enhancement. An edge-preserving filter was also applied post-CLAHE to maintain the sharpness of the text edges without introducing artifacts.

3.5.2 Text Region-Based Rotation

In our preprocessing framework, we have implemented fine-tuning rotation through text region alignment to ensure accurate text detection and extraction from exam scripts. We have adopted a two-step correction process. Initially, we have employed a standard skew detection algorithm to estimate the skew angle θ of the document, represented by the transformation matrix:

$$T(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.3)$$

This step provides an initial rotation to correct for skew. Following this, we have introduced a novel refinement step that utilizes text region analysis. By applying a pretrained OCR model to identify text areas within the document, we have computed the average

angle of the detected bounding boxes as follows:

$$\text{Angle}_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N \text{atan2}(h_i, w_i) \quad (3.4)$$

where h_i and w_i denote the height and width of the i^{th} detected bounding box, respectively. The function $\text{atan2}(h_i, w_i)$ is an arctangent function that calculates the angle of a point with respect to the origin in a 2D Cartesian coordinate system considering both the horizontal and vertical components. This enables us to refine the initial skew correction by adjusting the rotation angle:

$$\theta_{\text{final}} = \theta + \text{Angle}_{\text{avg}} \quad (3.5)$$

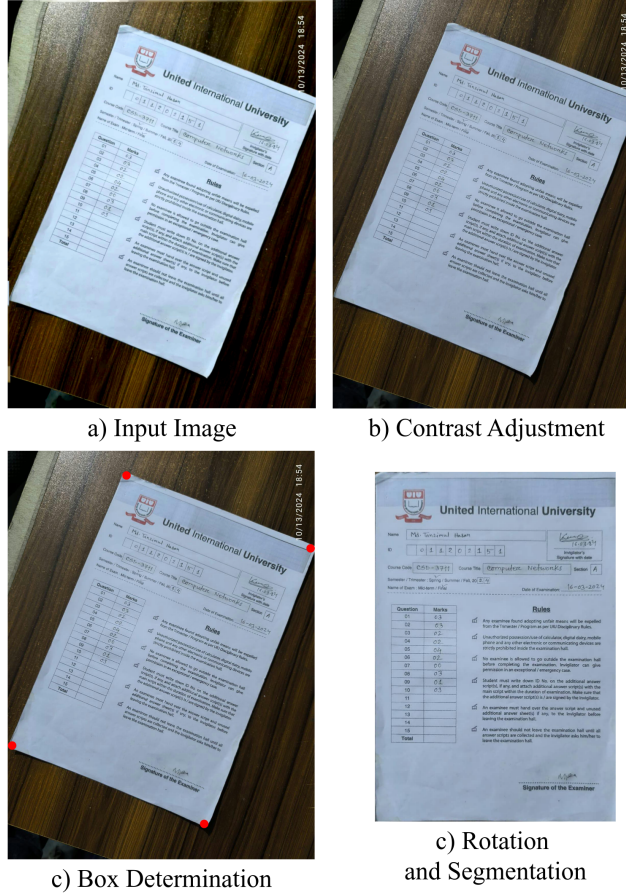


Figure 3.3: Background segmentation and dynamic rotation handling output

Ultimately, the method dynamically adapts to the unique layout of each document by considering the spatial distribution and orientation of text regions.

3.5.3 Image Denoising

BM3D (Block-Matching and 3D Filtering) [43] serves as the foundational denoising technique employed in our preprocessing phase to enhance the quality of scanned exam script images. The primary objective of using BM3D is to effectively reduce noise while preserving the essential details for accurate character recognition tasks of the handwritten text. Our implementation modifies the existing BM3D approach in two significant ways to better suit the unique characteristics of exam scripts. Initially, we have adapted the block size and overlap parameters based on the specific dimensions and features of handwritten text. We have ensured that variations in stroke width and character spacing are preserved during the denoising process by using smaller blocks with increased overlap. This adaptation allows for the more accurate matching of similar blocks. For m and n to be the dimensions of each block, the noisy image Z is initially divided into overlapping blocks Z_i :

$$Z_i = Z[x : x + m, y : y + n] \quad (3.6)$$

Following this, we have tailored the similarity metrics used in the block-matching step. Instead of relying solely on pixel-wise differences, we have incorporated a feature-based similarity measure that accounts for the structural characteristics of handwriting by prioritizing the matching blocks that exhibit similar local patterns, such as curves and angles. For each block Z_i , the similar blocks Z_j are computed based on our modified similarity metric as:

$$S(Z_i, Z_j) = \|f(Z_i) - f(Z_j)\|_2^2 \quad (3.7)$$

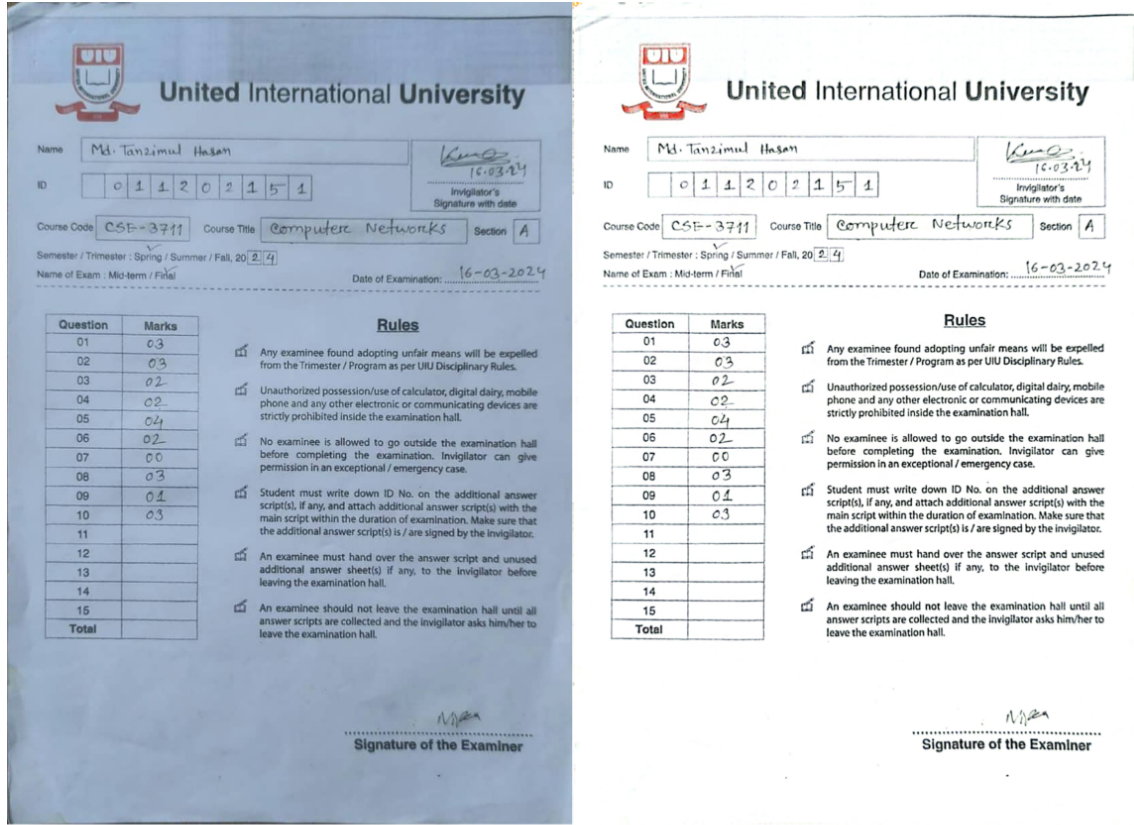
where f represents a feature extraction function that captures the structural properties of the handwritten text. The 3D collaborative filtering step is then performed which results in an estimated denoised block Y . The overall denoising process is expressed as:

$$\hat{Y} = \underset{Y}{\operatorname{argmin}} \left\{ \sum_i \|Z_i - Y\|_2^2 + \lambda \|Y\|_1 \right\} \quad (3.8)$$

where λ is a regularization parameter that balances fidelity to the noisy data against sparsity in the solution. Finally, the overall denoised image has been reconstructed from the individual denoised blocks. Fig. 3.4 illustrates the BM3D filtering output.

3.5.4 Dataset Augmentation

In our study, we have leveraged a Generative Adversarial Network (GAN) based augmentation technique instead of traditional augmentation methods like cropping, zooming, or rotating. Conventional augmentation techniques, while useful for many image processing tasks, pose significant risks for our dataset. Any alteration such as cropping, zooming, or rotating can result in the loss or distortion of the content in the exam scripts, such as student details, course information, and question-mark table. Given the need to preserve these documents' exact layouts and details, the GAN has been implemented to generate



a) Segmented Image

b) BM3D Filtered Image

Figure 3.4: Image output before and after BM3D filtering

synthetic exam script images. GANs operate by having two neural networks—the generator and the discriminator—compete against each other. The generator creates new, synthetic images, while the discriminator evaluates whether an image is real (from the dataset) or fake (generated). The generator learns to produce highly realistic images that closely resemble the original scripts through this adversarial process [44]. This approach allowed us to expand our dataset with diverse, yet structurally consistent, images, improving the training of our OCR models without compromising the content integrity.

3.6 Segmentation

The segmentation of exam scripts into distinct sections is achieved through a comprehensive pipeline that incorporates techniques such as background isolation, upper and lower section separation, label detection, and table segmentation. The segmentation process for exam script images is outlined in Algorithm 2. Detailed explanations of each step are provided in the subsequent sections.

Table 3.1: Tasks Assignment

Task ID	Assignee	Start Date	End Date
1.1	A. Rahman, Umar, Tanzim	Jan 25	Feb 23
1.2	A. Rahman, Tanzim	Jan 26	Feb 20
1.3	Tanzim, A. Kader	Jan 29	Feb 27
2.1	Umar, A. Rahman	Jan 28	Feb 19
2.2	A. Rahman	Feb 22	March 9
2.3	Tanzim, Umar	Feb 25	March 18
3.1	Umar, A. Rahman, Tanzim, A. Kader	Feb 28	March 30
3.2	A. Kader, A. Rahman	Feb 22	March 27
3.3	A. Rahman, Tanzim, Umar	March 3	April 15
4.1	A. Rahman, Tanzim	March 9	April 22
4.2	Umar	March 13	April 27
4.3	A. Kader, Tanzim	March 24	April 29
5.1	A. Rahman	April 17	June 13
5.2	Tanzim, A. Rahman, Umar, A. Kader	April 6	May 26
5.3	Tanzim	May 21	June 19
6.1	A. Kader, Tanzim, A. Rahman, Umar	May 29	June 30
6.2	A. Rahman	July 2	Sep 4
6.3	Umar	June 26	Aug 30
7.1	Umar, Tanzim	Aug 29	Sep 29
7.2	A. Kader, Tanzim, A. Rahman, Umar	sep 14	Oct 27
7.3	A. Rahman	Aug 24	Oct 15
8.1	Tanzim, A. Kader	Feb 9	Oct 30
8.2	Umar, A. Rahman	March 2	Oct 31

Table 3.2: Dataset distribution

Aspects	Original	Augmented	Final (Processed)
Type	Page Image	Page Image	Page Image
Format	Variable	PNG	PNG
Dimension	Variable	256×256	256×256
Size	412	300	712

Algorithm 2 Segmentation pipeline

```

1: Input: Preprocessed Exam Script Image  $I$ 
2:
3: Step 1: Script Segmentation
4:  $I_{\text{gray}} \leftarrow \text{convert\_to\_grayscale}(I)$ 
5:  $I_{\text{binary}} \leftarrow \text{otsu\_threshold}(I_{\text{gray}})$ 
6:  $C \leftarrow \text{extract\_contours}(I_{\text{binary}})$ 
7:  $I_{\text{script}} \leftarrow \text{crop\_to\_largest\_contour}(C)$ 
8:
9: Step 2: Section Separation
10:  $I_{\text{edges}} \leftarrow \text{canny\_edges}(I_{\text{script}})$ 
11:  $B \leftarrow \text{detect\_table\_boundary}(I_{\text{edges}})$ 
12:  $I_{\text{upper}}, I_{\text{lower}} \leftarrow \text{split\_image}(I_{\text{script}}, B)$ 
13:
14: Step 3: Upper Section Processing
15:  $F \leftarrow \text{ocr\_detect\_labels}(I_{\text{upper}})$ 
16: for all label  $L_i$  in  $F$  do
17:    $H_i \leftarrow \text{extract\_handwritten\_data}(I_{\text{upper}}, L_i)$ 
18: end for
19:
20: Step 4: Lower Section Processing
21:  $C_{\text{table}} \leftarrow \text{detect\_table\_contours}(I_{\text{lower}})$ 
22:  $E \leftarrow \text{segment\_entries}(C_{\text{table}}, I_{\text{lower}})$ 
23:
24: Output: Segmented Image  $I_{\text{segmented}}$ , Text Fields  $F$ , Table Entries  $E$ 

```

3.6.1 Background Segmentation

To ensure the focus on the exam script, the first step involves isolating the page from any extraneous background captured in the image since images contain unwanted elements such as table surfaces, surrounding papers, shadows, etc. Initially, the input image is converted to grayscale using a weighted sum approach, which simplifies subsequent processing by reducing the image to a single channel.

Next, Otsu's thresholding is applied to binarize the image by selecting an optimal threshold T that maximizes the variance between foreground (script) and background classes:

$$T = \arg \max_{\tau} \left(\frac{n_1(\tau)n_2(\tau)}{n^2} (\mu_1(\tau) - \mu_2(\tau))^2 \right) \quad (3.9)$$

where $n_1(\tau)$ and $n_2(\tau)$ denote the number of pixels in the foreground and background classes, respectively, and $\mu_1(\tau)$ and $\mu_2(\tau)$ are their corresponding mean intensities. The optimization over τ yields a threshold T that distinguishes the script from the background with the highest class separability.

The resulting binary image highlights the script against the background. We have then identified contours extracting a set of contours $C = \{c_1, c_2, \dots, c_n\}$. The largest contour, assumed to correspond to the exam script, is selected, and a bounding box is generated

to crop the image, effectively removing any non-script background elements.

3.6.2 Scripts' Part Separation

After isolating the script from the background, the next step involves dividing the page into upper and lower sections containing student-course and question-mark information, respectively, as each section requires a distinct processing approach. We have detected the location of the marks table by applying edge detection using the canny operator, followed by a Hough Line Transform [45] to identify horizontal and vertical lines. The lines represent potential table borders allowing us to pinpoint the table's top boundary. The image is then split at this boundary, with the upper part containing textual information and the lower part containing the marks table.

3.6.3 Labels and Text Fields Detection

The upper part (I_{upper}) of the exam script contains printed labels alongside their corresponding handwritten entries, which include the student's name, ID, and course code. To effectively segment these fields, we have detected and recognized the textual labels using an OCR engine which resulted in a set of labels $L = \{L_1, L_2, \dots, L_i\}$. For each detected label L_i , we have identified the corresponding handwritten data H_i positioned in a box placeholder adjacent to the label as:

$$H_i = I_{upper}[y_i : y_i + h_i, x_i + w_i : x_i + w_i + w_{\text{field}}] \quad (3.10)$$

where w_{field} denotes the width of the box designated for the handwritten entry, while x_i and y_i represent the coordinates of label L_i , and w_i and h_i indicate its width and height, respectively. Fig. 3.5 shows the information extraction output.

The figure displays four examples of exam script forms from United International University. Each form contains the following fields and extracted information:

- Name:** Hasan Anik (top-left), Dhiman Mozumder (top-right), Hasan Anik (bottom-left), Dhiman Mozumder (bottom-right).
- ID:** 011202227 (top-left), 011202228 (top-right), 011202227 (bottom-left), 011202228 (bottom-right).
- Course Code:** CSE-4611 (all four).
- Course Title:** Compiler Design (all four).
- Section:** B (all four).
- Semester / Trimester:** Spring / Summer / Fall, 2024 (all four).
- Name of Exam:** Mid-term / Final (all four).
- Date of Examination:** 10-03-2024 (all four).

Figure 3.5: Text fields localization and information extraction

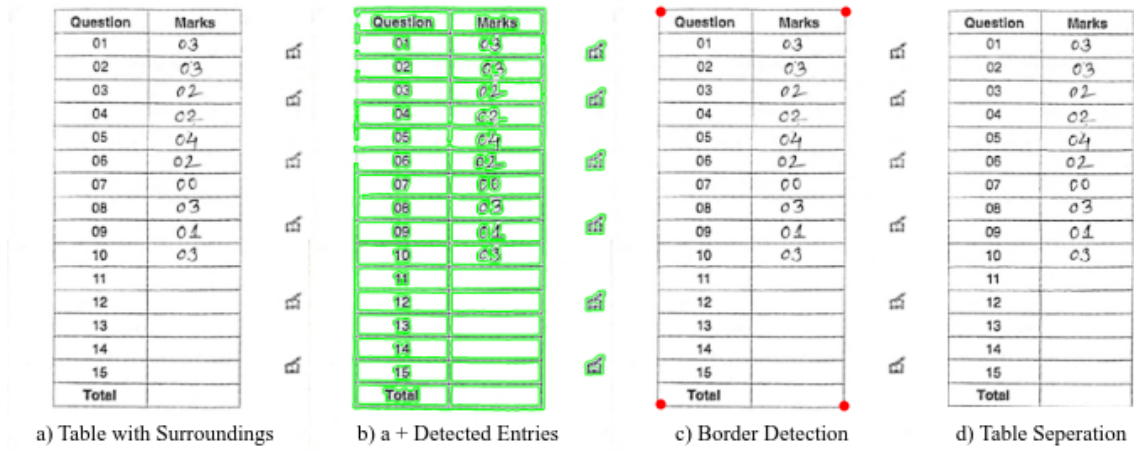


Figure 3.6: Contour detection and table segmentation

3.6.4 Table Detection and Entries Segmentation

Accurate segmentation of the table data from the lower part of the image is essential for mapping each question to its corresponding marks. The process begins with the segmentation of the table structure. We have initiated it by applying edge detection techniques to identify the table contours. The lower portion of the image, denoted as I_l , is first converted into a binary image B_l using thresholding. The gradient of the image is computed as follows:

$$\nabla I_l = \left(\frac{\partial I_l}{\partial x}, \frac{\partial I_l}{\partial y} \right) \quad (3.11)$$

Then defined by the set of points where the gradient magnitude exceeds a specified threshold T , the contours C are then detected as:

$$C = \{(x, y) \mid \|\nabla I_l(x, y)\| > T\} \quad (3.12)$$

Upon identifying the contours, as seen in Fig. 3.6, we have focused on isolating the table by removing extraneous components that do not form part of the table structure. Morphological operations, including dilation and erosion, are applied to refine the binary image B_l and enhance the contours. The processed images, B_{clean} , are generated as:

$$B_{clean}(x, y) = (B_l \ominus E) \oplus D \quad (3.13)$$

where E and D represent the structuring elements for erosion and dilation, respectively. Following the isolation of the table, we have proceeded to segment individual entries corresponding to questions and their respective marks. This segmentation utilizes the existing binary image B_{clean} to delineate the individual cells. Connected component analysis is then employed to identify and extract individual text entries. Each connected component

Question	Marks	Question	Marks	Question	Marks
01	03	01	03	01	03
02	03	02	03	02	03
03	02	03	02	03	02
04	02	04	02	04	02
05	04	05	04	05	04
06	02	06	02	06	02
07	00	07	00	07	00
08	03	08	03	08	03
09	01	09	01	09	01
10	03	10	03	10	03
11		11		11	
12		12		12	
13		13		13	
14		14		14	
15		15		15	
Total		Total		Total	

Figure 3.7: Segmentation of question-mark entries from the table

C_{comp} is defined as:

$$C_{comp} = \{(x, y) \mid B_{clean}(x, y) = 1 \text{ and connected}\} \quad (3.14)$$

Fig. 3.7 illustrates the outcome which ensures that each question number and its corresponding marks are distinctly identified forming question-mark pairs ready for subsequent processing

3.7 Text Processing

The text processing of exam scripts is carried out through a pipeline that incorporates techniques such as feature extraction with an ensemble of OCR models, majority voting for enhanced character recognition, post-processing for accuracy improvement, and validation of structured fields. The text processing pipeline is outlined in Algorithm 3, with detailed explanations of each step provided in the subsequent sections.

Algorithm 3 Text processing pipeline

```

1: Input: Preprocessed Exam Script Images  $I'$ 
2:
3: Step 1: Text Extraction
4: for all image  $i'$  in  $I'$  do
5:    $X \leftarrow \text{extract\_feature\_maps}(i')$ 
6:    $Y \leftarrow \text{recognize\_characters}(X)$  ▷ using ensemble
7: end for
8:
9: Step 2: Majority Voting
10: for all character  $y_t$  at timestep  $t$  do
11:    $\hat{y}_t \leftarrow \text{mode}\{y_t^{M_1}, y_t^{M_2}, \dots, y_t^{M_n}\}$ 
12: end for
13:
14: Step 3: Post-Processing
15: for all extracted character  $y_t$  do
16:    $\omega \leftarrow \text{calculate\_confidence}(y_t)$ 
17:   if  $\omega < 0.5$  then
18:      $y_t \leftarrow \text{correct\_recognition\_error}(y_t)$ 
19:   end if
20: end for
21:
22: Step 4: Validation of Structured Fields
23: for all field  $f$  in extracted text do
24:   if  $f$  matches pattern then
25:      $f \leftarrow \text{validate\_format}(f)$ 
26:   end if
27: end for
28:
29: Step 5: Output Text Formation
30:  $E \leftarrow \text{tokenize\_extracted\_text}(Y)$ 
31:  $O \leftarrow \text{transform\_to\_structured\_format}(E)$ 
32:
33: Output: Formatted Output  $O$ 

```

Text Extraction

Our text extraction pipeline is implemented as an ensemble framework that leverages a combination of four state-of-the-art OCR models: Tesseract [46], EasyOCR [47], CRAFT [48], and TrOCR [49] which are designed to handle both printed and handwritten text under varied conditions, including noise, skew, and irregular layouts. We have utilized this ensemble approach to extract text from exam scripts' upper and lower sections. The process begins with detecting text regions across the image, where convolutional operations are applied to extract feature maps X from the input image to identify printed and handwritten characters. Upon identifying text regions, we have employed a recurrent neural network architecture to recognize the character sequences. The character recogni-

tion models process these feature maps using bidirectional LSTMs to capture contextual information within sequences. The output sequence of characters $Y = \{y_1, y_2, \dots, y_T\}$ is generated by maximizing the conditional probability $P(Y|X)$ of character sequences given the image features:

$$P(Y|X) = \prod_{t=1}^T P(y_t|X) \quad (3.15)$$

This ensures that the sequential relationships in handwritten and printed text are captured effectively, regardless of text layout or distortion.

To ensure accurate text extraction, the final step aggregates the results of all the models using a majority voting system. For each character y_t at timestep t , the output is determined by selecting the most frequent prediction from the ensemble of models as:

$$\hat{y}_t = \text{mode}\{y_t^{M_1}, y_t^{M_2}, \dots, y_t^{M_n}\} \quad (3.16)$$

3.7.1 Post-Processing

To enhance the reliability of extracted information and reduce OCR errors and inconsistencies across different models, we have implemented a robust ensemble-based post-processing strategy. This process starts by assigning confidence scores $\omega \in [0, 1]$ to each extracted character y_t . The confidence score is calculated as the weighted average of character probabilities from all models, defined as:

$$\omega = \frac{1}{M} \sum_{m=1}^M \frac{P_m(y_t)}{TC_m} \quad (3.17)$$

where $P_m(y_t)$ is the probability of character y_t predicted by model m , TC_m is the total number of characters detected by model m , and M is the total number of models in the ensemble. This score helps detect low-confidence characters ($\omega < 0.5$), which are likely to have been misrecognized.

We have incorporated a format-based validation to automatically correct recognition errors for fields with predefined patterns, such as student IDs and course codes. For instance, student IDs follow a fixed format of 9 to 10 digits (depending on the admitted semester), which are validated using regular expression matching. If the recognized text deviates from this expected pattern, we have applied corrective measures based on the candidate with the highest probability of being chosen from the ensemble output.

For the marks table, where numerical values are handwritten, we have ensured consistency by validating the sum of individual question scores against the total marks encoded in the filename. The total marks T_{marks} are computed as:

$$T_{marks} = \sum_{i=1}^N \sum_{j=1}^K M_{ij} \quad (3.18)$$

where M_{ij} is the mark for question j in exam script i , N is the total number of scripts, and K represents the total number of questions.

3.7.2 Output Text Formation

In the document formation phase, we have utilized a Transformer-based architecture to format the output from the ensemble of OCR models into structured text. The process begins with tokenizing the extracted character sequences into embeddings $E = \{e_1, e_2, \dots, e_n\}$, where each character c_i is represented as a vector $e_i = \text{Embed}(c_i)$. Positional encoding is then applied to incorporate sequence order, resulting in $E_{pos} = E + P$, where P denotes the positional vectors that provide information about the position of each token in the sequence.

This sequence of embeddings is passed through the Transformer's multi-head attention mechanism. The core processing occurs in the Transformer attention block, where we have computed scaled dot-product attention to allow the model to focus on relevant tokens dynamically. The attention block is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3.19)$$

where Q , K , and V represent the query, key, and value matrices, respectively, and d_k is the dimensionality of the key vectors, which helps in scaling the dot product to prevent large values that might lead to vanishing gradients. This attention mechanism allows the model to learn contextual relationships and capture local and global dependencies within the text sequence. The output from the attention layer is passed through a feed-forward network (FFN) that transforms the embeddings as follows:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (3.20)$$

where W_1 and W_2 are learned weight matrices, and b_1 and b_2 are bias terms. This network refines the embeddings and prepares the sequence for final prediction. The output is then passed through a linear layer followed by a Softmax activation, which produces a probability distribution over possible characters for each position in the sequence:

$$Y = \text{Softmax}(W_{final}H + b_{final}) \quad (3.21)$$

where H represents the hidden states from the final feed-forward layer, W_{final} is the learned weight matrix, and b_{final} is the bias term. Finally, to ensure the integrity of the formatted document, we have validated the output against expected patterns using regular expressions.

Chapter 4

Implementation and Results

This chapter presents the implementation details and results of the project, covering the environmental setup, evaluation criteria, training analysis, result validation, comparisons with existing methods, and findings from the ablation study.

4.1 Environment Setup

The experiments are conducted in a Google Colaboratory environment using an NVIDIA T4 GPU with 15GB of GPU memory, supported by 12.7GB of system RAM and 112.6GB of available disk space. The methodology has been evaluated based on its ability to correctly extract the four key fields from the exam scripts. Experimental analysis with different configurations is discussed in the ablation study in subsection 4.7.

4.2 Evaluation Criteria

To evaluate the effectiveness of our system, we have employed several performance metrics: accuracy, precision, recall, F1-Score, and error rate. Accuracy measures the overall correctness, precision indicates the ratio of correctly identified fields to all identified fields, recall measures the system’s ability to find all relevant fields, F1-Score balances precision and recall to account for their trade-off, and error rates represent the proportion of incorrect predictions made by the model compared to the total number of predictions. These metrics provide a comprehensive view of the system’s performance in recognizing key fields such as student name, student ID, and total marks from the test data. The evaluation is performed on ground truth values provided in the format $\langle \text{Name} \rangle \langle \text{ID} \rangle \langle \text{Total Marks} \rangle$. It is worth noting that we lack the ground truth for each mark corresponding to its respective questions in our collected dataset, thus for the total mark entries evaluation, we have validated only the sum of the extracted marks against the total marks provided in the filename. The evaluation metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (4.1)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4.2)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4.3)$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.4)$$

$$\text{Error rate} = 1 - \text{Accuracy} \quad (4.5)$$

where TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative cases, respectively.

4.3 Training Analysis

The training process for our ensemble OCR model is carried out over 60 epochs, using images resized to 256×256 pixels. We have used a batch size of 16 for efficient training, with a learning rate set to 0.0001 for stable convergence. The convolutional layers use a kernel size of 3×3 . The dataset, consisting of 712 images, is split into 80% for training (570 images) and 20% for testing (142 images). The Adam optimizer is used for efficient gradient-based optimization throughout the training process.

As seen in Fig. 4.1, our ensemble model, integrating Tesseract, EasyOCR, CRAFT, and TrOCR through a majority-voting mechanism, progressively improved over the epochs. The best test accuracy of 99.32% is recorded at the 58th epoch, accompanied by a validation accuracy of 99.01%. On the other hand, the lowest training loss of 0.0116 and validation loss of 0.0192 is achieved in the 60th epoch. Fig. 4.1 shows the loss progression over the epochs for training and validation tasks.

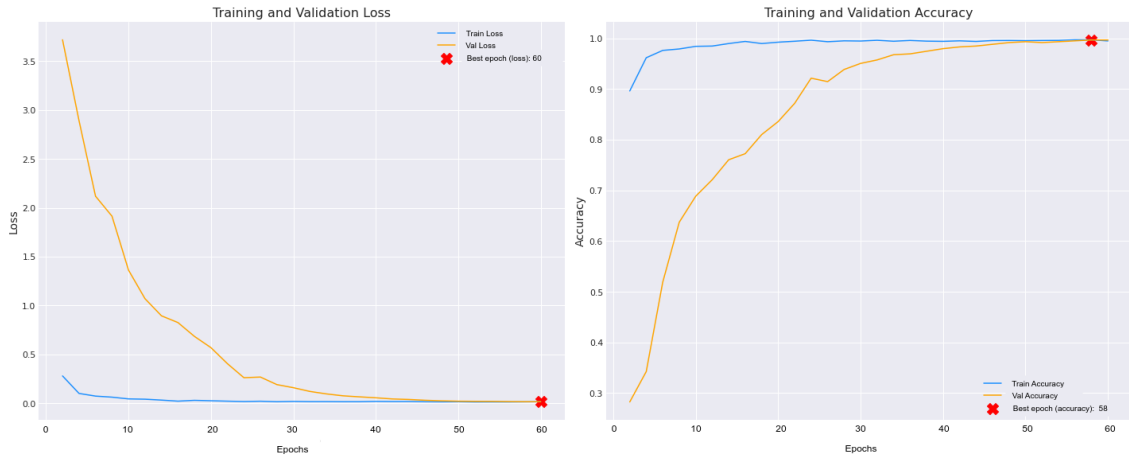


Figure 4.1: Training and validation accuracy and loss over epochs

4.4 Information Validation

In this section, we have evaluated our framework’s performance in detecting, segmenting, and recognizing the name, ID, and course code fields from exam script images using performance metrics such as accuracy, precision, recall, F1-Score, and error rate. The model has achieved exceptional performance in name entry detection, with an accuracy of 97.89%, a precision of 98.58%, and a recall of 99.29%, resulting in a low error rate of 2.11%. Furthermore, the segmentation accuracy for names stands at 97.08%, along with a precision of 97.74% and recall of 99.24%, which highlights its effectiveness in accurately isolating name entries even amid varying handwriting styles. For ID entry detection, the model maintains robust performance achieving an accuracy of 98.59%, with precision and recall both recorded at 99.26% yielding a minimal error rate of 1.41%. This high level of accuracy shows the model’s strong reliability in recognizing structured fields, which is further supported by a segmentation accuracy of 97.18% which indicates effective extraction practices in a standardized format. Course code detection also yields promising results, with an accuracy of 96.48% and a precision of 97.81%, coupled with a segmentation accuracy of 97.89%. The overall validation results of our model excel in detection and segmentation tasks, consistently achieving metrics above 96%. Although the OCR performance for names reveals a slight decline at 95.77%, the model maintains high precision and recall rates thus indicating that it successfully extracts information in most scenarios despite the inherent challenges of handwriting variability. Table 4.1 shows the detailed result for information validation.

Table 4.1: Information validation performance metrics

Field	Tasks	Accuracy	Precision	Recall	F1-Score	Error Rate
<i>Name</i>	Detection	97.89%	98.58%	99.29%	98.93%	2.11%
	Text Field Segmentation	97.08%	97.74%	99.24%	98.48%	2.92%
	OCR Validation	95.77%	97.10%	98.53%	97.81%	4.23%
<i>ID</i>	Detection	98.59%	99.26%	99.26%	99.26%	1.41%
	Text Field Segmentation	97.18%	99.25%	97.79%	98.52%	2.82%
	OCR Validation	96.48%	97.10%	99.26%	98.17%	3.52%
<i>Course Code</i>	Detection	96.48%	97.81%	98.53%	98.17%	3.52%
	Text Field Segmentation	97.89%	99.27%	98.55%	98.91%	2.11%
<i>Table</i>	Detection & Segmentation	97.18%	99.28%	97.86%	98.56%	2.82%
<i>Entries</i>	Detection & Segmentation	94.37%	95.59%	98.48%	97.01%	5.63%
	Total Marks Validation	95.07%	96.38%	98.52%	97.44%	4.93%

4.5 Mark Evaluation

In this section, we have evaluated the performance of our system in detecting and extracting marks assigned to questions from exam scripts focusing on two important tasks: the segmentation of the marks table, and the extraction of marks for each question via OCR. We have also validated the total marks obtained against the ground truth (GT) for each

exam script. Table 4.1 shows the performance metrics for question-mark evaluation. The results for table segmentation reveal an accuracy of 97.18%, with remarkable precision of 99.28% and recall of 97.86% leading to an F1-Score of 98.56% and an error rate of 2.82%. These metrics indicate that our system effectively identifies the table structure which is well-defined and distinct from other elements of the exam script. While evaluating the extraction of individual marks, the system achieves an accuracy of 94.37%, with a precision of 95.59%, a recall of 98.48%, and an F1-Score of 97.01%. The error rate here is slightly higher at 5.63%, which can be attributed to challenges in recognizing handwritten marks due to variations in stroke quality and spacing. In the total marks validation task, where the sum of all marks is compared against the ground truth, our model achieves an accuracy of 95.07%, a precision of 96.38%, a recall of 98.52%, and an F1-Score of 97.44%, with an error rate of 4.93%. This indicates that the model effectively aggregates the individual marks and closely aligns with the ground truth. Our findings demonstrate that our framework excels in both segmentation and extraction tasks consistently achieving performance metrics above 94% across all evaluated fields.

4.6 Comparison with State-of-the-Art Models

To evaluate the generalizability of our proposed framework, we conducted experiments using the ICDAR2013 dataset. As shown in Table 4.2, our model (uFOIL) achieves competitive results across all metrics for table detection, with 95.89% precision, 97.86% recall, and 96.87% F1-score. In comparison to CDeC-Net, which recorded 95.3% precision and 93.4% recall, uFOIL demonstrates a clear improvement in recall. Although LGPMA achieved a slightly higher recall at 99.1%, our model maintains a more balanced performance across all metrics, with only a minimal difference in recall while outperforming in precision. Compared to TabCellNet and TableNet, which have lower precision scores of 89.2% and 93.07%, respectively, uFOIL shows stronger precision, while also maintaining robust recall and F1-score results

Table 4.2: Comparison with the state-of-the-art models

Reference	Precision	Recall	F1-Score	Dataset
CDeC-Net [21]	95.30%	93.40%	94.40%	ICDAR2019
LGPMA [24]	96.70%	99.10%	97.90%	ICDAR2013
TableNet [22]	93.07%	90.01%	91.51%	ICDAR2013
TabCellNet [31]	89.20%	98.7	93.71%	ICDAR2013
uFOIL (Ours)	95.89%	97.86%	96.87%	ICDAR2013

4.7 Ablation Study

In this section, we have conducted an ablation study to assess the impact of different configurations on the performance of our proposed system. We have evaluated individual models—Tesseract, EasyOCR, CRAFT, and TrOCR—alongside their ensemble performance in validating name and ID entries, as summarized in Table 4.3. Tesseract and TrOCR emerge as the top performers, with Tesseract achieving an accuracy of 93.55% for name entries, supported by a precision of 94.71% and a recall of 95.47%. However, Tesseract also has an error rate of 6.45%, a character error rate of 5.55%, and a word error rate of 10.81%. On the other hand, EasyOCR shows the weakest overall performance, with an accuracy of 88.94% for names and an error rate of 11.06%. CRAFT performs moderately well with an accuracy of 92.20%, while TrOCR achieves the highest performance among the individual models, with an accuracy of 94.75%. The proposed ensemble model combines outputs from four OCR systems using majority voting achieving an accuracy of 95.77%, and 96.48% for Name, and ID, respectively. It has a precision of 97.10%, a recall of 98.53%, and an F1-score of 97.81%. The error rate is reduced to 4.23%, with a character error rate of 2.57% and a word error rate of 5.26%.

Table 4.3: Performance analysis: Name, and ID

Model	Field	A ↑	P ↑	R ↑	F1 ↑	ER ↓	CER ↓	WER ↓	IT ↓	ACM (1) ↑	ACM (2) ↑	ACM (3) ↑
<i>Tesseract</i>	Name	93.55%	94.71%	95.47%	95.09%	6.45%	5.55%	10.81%	0.9s	95.13%	96.52%	97.33%
	ID	92.08%	94.12%	96.31%	95.19%	7.92%	4.23%	8.54%	0.7s	96.22%	97.02%	97.85%
<i>EasyOCR</i>	Name	88.94%	91.50%	90.04%	90.77%	11.06%	4.84%	9.66%	1.2s	92.70%	94.12%	94.94%
	ID	89.67%	90.94%	90.06%	90.50%	10.33%	3.82%	7.50%	1.0s	92.80%	95.12%	96.54%
<i>CRAFT</i>	Name	92.20%	90.67%	91.11%	90.89%	7.80%	6.16%	11.56%	2.1s	93.23%	95.60%	96.32%
	ID	93.50%	92.14%	94.30%	93.21%	6.50%	5.37%	9.28%	1.9s	94.33%	96.74%	97.07%
<i>TrOCR</i>	Name	94.75%	94.38%	96.64%	95.49%	5.25%	4.14%	7.96%	1.5s	94.95%	96.32%	97.56%
	ID	93.09%	93.41%	95.48%	94.43%	6.91%	3.63%	6.58%	1.3s	93.93%	95.20%	96.83%
<i>Ensemble</i>	Name	95.77%	97.10%	98.53%	97.81%	4.23%	2.57%	5.26%	2.7s	97.90%	98.75%	99.30%
	ID	96.48%	97.10%	99.26%	98.17%	3.52%	1.88%	3.83%	2.3s	98.10%	99.10%	99.45%

The abbreviations used in the table are as follows: A: Accuracy, P: Precision, R: Recall, F1: F1-Score, ER: Error Rate, CER: Character Error Rate, WER: Word Error Rate, IT: Inference Time (in seconds), and ACM (1/2/3): Accuracy with n Character Mismatch tolerances. The best results are highlighted in bold font.

Table 4.4 further explores the total marks validation performance. Tesseract achieves

an accuracy of 93.45%, with precision, and recall of 93.60%, and 94.25%, respectively. EasyOCR records lower performance, with an accuracy of 89.72% and a higher error rate of 10.28%. On the other hand, CRAFT, and TrOCR perform better achieving accuracies of 90.03%, and 92.25%, respectively, but still do not reach the reliability of the ensemble model. The ensemble excels in this task, achieving an accuracy of 95.07%, precision of 96.38%, and recall of 98.52%, with an F1-score of 97.44% and a low error rate of 4.93%. Additionally, while evaluating the accuracy of the models with character mismatches, the ensemble shows significant improvement. The ensemble model achieves 97.90% accuracy when allowing for one character mismatch, 98.75% with two mismatches, and 99.30% with three mismatches.

Table 4.4: Performance analysis: Question-Mark validation

Model	Accuracy	Precision	Recall	F1-Score	ER ↓	CER ↓	WER ↓	IT ↓	ANM 1 (%)	ANM 2 (%)	ANM 3 (%)
Tesseract	93.45%	93.60%	94.25%	93.92%	6.55%	5.32%	9.83%	0.8s	94.21%	97.55%	97.82%
EasyOCR	89.72%	90.51%	91.41%	90.95%	10.28%	8.54%	13.21%	1.2s	90.47%	93.22%	94.70%
CRAFT	90.03%	89.81%	90.53%	90.17%	9.97%	7.12%	11.94%	1.3s	94.04%	94.93%	96.07%
TrOCR	92.25%	93.72%	92.64%	93.17%	7.75%	4.56%	10.53%	1.0s	93.54%	96.08%	96.94%
Ensemble	95.07%	96.38%	98.52%	97.44%	4.93%	2.51%	5.23%	2.7s	97.10%	98.91%	99.05%

In this table, ANM (1/2/3) refers to the accuracy achieved with n number of mismatch tolerances during the evaluation of individual entries (questions and marks). The best results are highlighted in bold font.

Chapter 5

Standards and Design Constraints

This chapter outlines the standards and design constraints that guide the development of the proposed framework. These standards ensure compliance with engineering best practices, while the design constraints shape the framework’s functionality, usability, and effectiveness.

5.1 Compliance with Standards

The model was developed with a strong focus on meeting industry standards across communication, hardware, and software domains. This compliance ensures the model’s reliability, efficiency, and ethical handling of sensitive academic information. The hardware requirements, communication protocols, and software architecture used in uFOIL follow relevant standards, which are outlined below.

5.1.1 Model Standards

For the design of our model, we followed these key standards:

- **ISO/IEC 25012 - Data Quality Model:** This standard focuses on maintaining high data quality by ensuring accuracy, completeness, and relevance. Since uFOIL processes sensitive academic documents, it is essential to keep data quality high for reliable results.
- **ISO/IEC 20546 - Big Data Reference Architecture:** Although uFOIL does not process large-scale data, this standard offers guidelines for handling data efficiently. It helps uFOIL manage academic data (e.g., exam scripts) with integrity, scalability, and efficiency.
- **IEEE 7003-2021 - Algorithmic Bias Considerations:** This guideline aims to reduce algorithmic bias, ensuring fair processing of student data. It guides the design of the ensemble model and transformer components to ensure equity and inclusivity across various document formats and handwriting styles.

- **ISO/IEC TR 24028 - AI Risk Management:** This standard provides recommendations for managing AI-related risks, such as explainability, accuracy, and reliability. Following these guidelines supports uFOIL’s need for high accuracy in recognizing student names, IDs, and grades, while also enhancing accountability.

5.1.2 Hardware Standards

For hardware, we followed these standards to ensure optimal performance and data security:

- **GPU and TPU Support:** The model supports CUDA-enabled GPUs (e.g., NVIDIA) and Tensor Processing Units (TPUs) to enable faster training and inference. Compatibility with popular frameworks like TensorFlow and PyTorch ensures optimized performance, which speeds up both development and deployment.
- **Data Security and Storage Standards:** Storage solutions for sensitive academic data, such as student IDs, names, and grades, follow industry standards for data encryption and access control. This approach prevents unauthorized access and protects privacy and confidentiality, especially when data is stored on external or cloud servers.
- **Redundant Storage and Recovery Protocols:** To protect against data loss, our model uses redundant storage solutions, including cloud-based storage and RAID configurations for on-premises setups. This redundancy ensures data integrity and reliability, which are essential during critical academic evaluation periods.

5.2 Design Constraints

This section outlines the main design constraints for the model, ensuring it operates effectively within economic, ethical, and safety considerations.

5.2.1 Economic Constraints

- **Budget Limitations:** Our approach prioritizes cost-effective solutions for training, deployment, and maintenance to keep development and operating expenses within budget.
- **Resource Allocation:** The model optimizes computational resources, especially when scaled up for large document processing during exam periods, to keep expenses low without compromising performance.

5.2.2 Ethical Constraints

- **Data Privacy and Security:** The dataset, containing sensitive exam information, was kept private and secure. Data integrity was maintained throughout processing

to ensure accuracy and confidentiality.

- **Fairness and Non-Discrimination:** To ensure fair treatment, the model avoids biases in data processing. Testing is conducted on diverse document formats and handwriting styles to maintain equity.

5.3 Cost Analysis

Table 5.1 details the comprehensive breakdown of the probable expenses during the implementation of our model. The substances listed include model-building resources, communication modules, hardware, miscellaneous components, employee wages, and maintenance costs. Each substance is allocated a specific amount, with corresponding percentages indicating its contribution to the total project cost.

Table 5.1: Cost Analysis

Serial	Components	Amount (BDT)	Ratio
1	High-Performance Server	150,000	22.06%
2	Model Design	50,000	7.35%
3	Model Licensing	20,000	2.94%
4	Dataset Collection	30,000	4.41%
5	Data Storage	20,000	2.94%
6	Model Training	50,000	7.35%
7	Employee Wages	180,000	26.47%
8	System Maintenance	60,000	8.82%
9	Miscellaneous Expenses	90,000	13.24%
Total		680,000	100%

5.4 Complex Engineering Problem

Through analysis, our project meets key criteria that classify it as a complex engineering problem.

5.4.1 Knowledge Profile

Our model addresses several aspects of the engineering knowledge profile:

- **WA1:** *Apply knowledge of mathematics, science, and engineering fundamentals to solve complex problems.* Our approach combines scientific principles, mathematical models, and engineering fundamentals to solve challenging issues in automated document processing.

- **WA2:** *Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Problem-solving* Our approach helps to develop a framework that accurately extracts critical information (such as student names, IDs, and total marks) from diverse academic documents while ensuring compliance with privacy regulations and maintaining high operational efficiency.
- **WA4:** *Conduct investigations using research-based knowledge and methods, including experiments, data analysis, and synthesis.* Our project involves systematic research and data analysis to understand and address complex problems. We use structured experiments, data interpretation, and information synthesis to reach valid conclusions.
- **WA5:** *Select and apply appropriate tools, including modeling and prediction, with an understanding of limitations.* We apply prediction and modeling tools to address engineering challenges, carefully considering each tool's limitations to ensure practical, viable solutions.
- **WA9:** *Work effectively as an individual and as a team member or leader in multi-disciplinary settings.* This project indicates our ability to work both independently and collaboratively in diverse teams, which contributed to its successful completion.

As seen in Table 5.2, Our project aligns with the Washington Accord's expectations for complex engineering analysis fulfilling criteria such as WA1, WA4, WA5, and WA9.

Table 5.2: Mapping with complex engineering Knowledge Profile.

WA1	WA2	WA4	WA5	WA9
Engineering Knowledge	Problem Analysis	Investigation	Modern Tool Usage	Individual and Team-work
✓	✓	✓	✓	✓

5.4.2 Complex Problem Solving

Our project demonstrates effective problem-solving in several ways:

- **WP1:** We use core engineering principles (WK3, WK4, WK5, WK6, WK8) to design a model grounded in fundamentals. This approach helps us solve challenging issues by applying a solid, principle-based framework.
- **WP3:** Our work involves creative analysis and abstract thinking to solve complex problems without clear solutions. We develop unique models tailored to handle these specific challenges.

- **WP4:** The project addresses uncommon issues, such as handling difficult handwriting, rotated and blurred images, and unstructured text. Tackling these challenges shows our model’s ability to manage rare and complex scenarios.
- **WP7:** Our project deals with high-level problems involving multiple components and sub-problems, requiring an understanding of their interrelationships.

Additionally, the project meets these criteria:

- **Engineering Design:** Our model applies engineering principles in a specialized practice area, emphasizing practical problem-solving.
- **Information Extraction:** Our model successfully extracts meaningful information from handwritten text, a complex form of unstructured data.

By meeting WP1, WP3, WP4, and WP7 (see Table 5.3), and integrating engineering design, information extraction, and cross-domain processes, our project highlights the analytical, problem-solving, and domain-specific knowledge necessary for tackling complex engineering challenges.

Table 5.3: Mapping with complex engineering problem-solving.

WP1	WP2	WP3	WP4	WP5	WP6	WP7
Dept of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent of Stakeholder Involvement	Inter-dependence
✓	X	✓	✓	X	X	✓

5.4.3 Engineering Activities

Our method has satisfied some portion of engineering Activities

- **EA1:** *Involve the use of diverse resources (people, money, equipment, materials, information, and technologies).* Our model requires the utilization of diverse resources to accomplish its objectives. This may include the involvement of personnel with various expertise, equipment, materials, and relevant information.
- **EA3:** *Involve creative use of engineering principles and research-based knowledge in novel ways.* Our method shows the creative application of engineering principles and research-based knowledge. It involves innovative approaches and techniques that go beyond conventional solutions.

- **EA5:** *Can extend beyond previous experiences by applying principles-based approaches.* Our model goes beyond previous experiences by employing principles-based approaches. It utilizes fundamental engineering principles and uses them to address complex challenges in new and innovative ways.

As seen in Table 5.4, our method satisfies the criteria EA1, EA3, and EA5 by utilizing diverse resources, showcasing the creative use of engineering principles and research-based knowledge, and extending beyond previous experiences through principles-based approaches.

Table 5.4: Mapping with complex Engineering Activities.

EA1	EA2	EA3	EA4	EA5
Range of resources	Level of Interaction	Innovation	Consequences for society and environment	Familiarity
✓	X	✓	X	✓

Chapter 6

Conclusion

6.1 Summary

This research proposes a comprehensive, ensemble-based unsupervised learning framework designed to automate the extraction and validation of key information from exam scripts. The methodology incorporates advanced techniques, including precise table detection for mark extraction, segmentation of the script into relevant fields, and an ensemble of Optical Character Recognition (OCR) systems to handle both handwritten and typed text. These methods are combined with a domain-specific approach that effectively addresses variations in layout, rotation, and document quality, offering a dynamic solution for processing diverse exam scripts without the need for annotated training data. Our framework utilizes individual OCR models—Tesseract, EasyOCR, CRAFT, and TrOCR—integrated through a majority voting ensemble model. This ensemble approach demonstrates superior performance across all metrics compared to the individual models. Specifically, the ensemble model achieves an accuracy of 95.77% for student names and 96.48% for IDs, significantly outperforming standalone systems. It excels in precision (97.10%) and recall (98.53%), while error rates such as the character error rate (CER) and word error rate (WER) are reduced to 2.57% and 5.26%, respectively. The ablation study on total marks validation further validates the effectiveness of our framework. The ensemble model consistently outperforms individual OCR models in validating extracted marks, achieving an accuracy of 95.07%. The model shows strong potential for automating student details and marks extraction in academic settings, as well as for structured data extraction in administrative processing, academic evaluations, and record digitization.

6.2 Limitations

In developing and testing the uFOIL framework, we faced several limitations. Variability in handwriting styles significantly challenged OCR accuracy. Cursive and stylized writing caused inconsistencies, even with an ensemble of models. Errors in OCR output, particularly in fields like student IDs and tabulated marks, tended to propagate through the

pipeline. This adversely affected NLP tasks and structured data validation.

The complexity of language patterns in handwritten responses posed another challenge. Responses often included informal phrasing, abbreviations, and unique shorthand. Our Transformer-based model struggled to interpret these accurately. Additionally, image quality varied due to lighting, shadows, and noise. This led to missing characters and text artifacts. While preprocessing steps like denoising improved some images, OCR quality still suffered, especially with lower-quality inputs.

We also encountered issues with missing or incomplete structured data entries. This made reliable validation of certain fields difficult. Finally, although we used GAN-based augmentation to address dataset constraints, our dataset is specific to one institution's exam formats. This limits the model's application to other institutions' documents without further validation on a broader range of layouts and formats.

6.3 Future Work

Building upon the current state of the model, we have identified several key areas to guide future improvements:

- 1. Adaptive OCR and Enhanced Preprocessing for Handwriting Diversity:** Given the challenges with handwriting variability, our next steps include investigating adaptive OCR techniques and exploring advanced preprocessing tailored to specific handwriting characteristics. Fine-tuning OCR models on handwriting-focused datasets or incorporating handwriting-specific OCR modules are promising directions we intend to pursue.
- 2. Integrating Error Correction Mechanisms:** To address error propagation, we aim to implement error correction strategies such as dictionary-based corrections, model ensembling, and contextual embeddings that can identify and correct common OCR errors. This will help minimize cascading errors.
- 3. Handling Missing Structured Data Entries:** To improve the accuracy of structured data extraction, we plan to explore probabilistic methods for identifying and filling missing data points in tables and other structured fields.
- 4. Expanding to a Multi-Institutional Dataset:** To enhance generalizability, we aim to expand our dataset to include exam scripts and documents from multiple institutions. Testing across varied document structures will allow us to validate the framework's adaptability and broaden its applicability to different educational contexts.

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